

SP250 Homework 3

1. Let a, b, c, d, m with $m \neq 0$ be integers. Suppose that $a \equiv b \pmod{m}$.
 - a) Prove that $ac \equiv bc \pmod{m}$. Use the definition of modulo.
 - b) Now suppose additionally that $c \equiv d \pmod{m}$. Prove that $ac \equiv bd \pmod{m}$.

Solution:

2. a) Prove that out of any 10 integers, one can pick two, whose difference is a multiple of 9.

b) Fanny picked several integers such that no two of them add up to a multiple of 9 and no two differ by a multiple of 9. What is the maximum number of integers that Fanny could have picked? Explain your answer.

Solution:

3. 35 people have gathered for a party. Some of them shook hands. Each person wrote down on a piece of paper the number of hands they shook. Could the sum of all the numbers be equal to 239? Prove your answer.

Solution:

4. 25 students live in a dorm, which is shaped as a 5×5 square. One day, simultaneously, some of them decide to switch rooms in such a way that every student would move to a room that is neighboring his or her own (neighboring rooms are rooms that **share a side**). Prove that at least one student will end up staying in their own room.

Solution:

5. Does there exist a 10-digit number, consisting only of digits 2,5 and 8, that is a perfect square?

Solution:

6. All 10 digits from 0 to 9 are written on the board in some order. Sergey wants to put +’s and -’s (9 total, one between each consecutive pair of digits) such that the result evaluates to 0. Show that Sergey will never be able to achieve his task, no matter what order the digits are written in and no matter what signs he uses.

Solution:
